

What a fast decision model buys an AI agent

Telemetry, benchmarks and paired sessions, from curiosity to configuration

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An AI agent makes many small decisions: which model should run this session, may this file be read without asking the expensive model, how hard should the model think. A “system 1” decision model answers such typed questions in a fraction of a second. We asked what one buys an agent harness in quality, cost and speed. A week of live telemetry (464,524 events) showed decisions cost 159 ms against 4.8 s for a host model call, but that the headline agreement figure mostly measured a labelling artefact. A preregistered judge benchmark found Jev, Typesafe’s decision model, the most accurate per dollar; no judge was useful on real read decisions. A paired campaign of 1,036 multi-turn sessions measured what routing a session to a cheaper model really costs: with Fable 5.1 as host, routing once per session to Sonnet cost $0.56\times$ the plain host (geometric mean of per-pair ratios); with Opus 5.5 as host every routing arm cost more. Two preregistered follow-ups showed that part of that saving is reasoning effort (Sonnet at medium $0.821\times$ its default; Fable at medium $0.860\times$). A counterfactual analysis then suggested, and a preregistered holdout study of 1,224 sessions confirmed, that a one-line rule matches the decision model at the session-start choice ($1.008\times$, equivalent). On fresh scenarios the frozen Fable configuration cost $0.581\times$ plain Fable; routing on Opus still lost ($1.255\times$); medium effort did not help Opus. We end with what to configure, per host model and per harness, and with what we have not yet measured.

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1 The question

Coding agents spend money in one place: calls to a large *host model*. Every turn re-sends the conversation, and a session of a dozen turns can cost several dollars. Between those calls sit many small choices. Should this session run on the expensive model or a cheaper one? May the agent read a file without asking the model first? How much should the model “think”? A *decision model* (“system 1”) answers such questions: it takes a short state and a typed question and returns a probability for each allowed answer, in tens to hundreds of milliseconds. Jev, by Typesafe, is one; GPT-6 Luna and Sol, Cloudflare’s Clef and Clef-Flash, and small local models (Qwen3, Laya) can play the same role.

Our question was simple, and we followed it where it went: *what does a decision model buy an agent harness in quality, cost and speed?*

Key idea

A decision model is only worth what its decisions change. It pays off when a cheap, typed decision changes an expensive action, and it adds nothing when a fixed rule makes the same choice. Most of this report is about telling those two cases apart with measurements rather than estimates.

How the story goes. We first looked at a week of live telemetry (Section 2). We then built tools to grade decision models against labelled cases and real traces (Section 3), raised the difficulty to full paired agent sessions (Section 4), and reached a plateau where the question became how to configure the model rather than which one to pick (Section 5). Section 6 says what to set, per host model and per harness; Section 7 says what we did not finish.

How to read the evidence labels. Results marked [preregistered] test hypotheses fixed before the data; [screen] marks exploratory results. Cost ratios are *geometric means* of per-pair ratios unless labelled otherwise, and a 95 % interval is given wherever one exists. Every number is generated from committed evidence files. The full detail of each study, with its own figures and its peer review, is in the companion technical report v2 (docs/papers/2026-10-02-paired-measurement/), cited here as “the companion report”.

2 What a week of telemetry showed

The fast-decisions bundle records every decision it makes, privately and locally. Before measuring anything new we read what was already there: 464,524 events in 4,470 session files, 171,609 of them in the first week. 398 sessions were real work; 1,136 were tests.

Decisions are fast. Figure 1 puts the decision calls next to the thing they try to save. A local Qwen3 model proposed a file read in 48 ms at the median; Jev judged a session’s starting model in 159 ms (p95 311 ms). A host model call took 4.8 s at the median and 26.9 s at p95. Speed is not the obstacle.

The 7.1 % that was not what it seemed. The telemetry reported that the read-shortcut proposals agreed with the host’s next action only 7.1 % of the time (117 of 1,642). Taken at face value, that condemns the idea. Decomposed, it says little about the decision model:

- 1,491 of the 1,642 decisions followed a host call to a tool the proposal could not contain, so they could never match. “Agreement” was partly a labelling artefact.
- 648 were scored by a synthetic demo backend that always picks the first candidate.

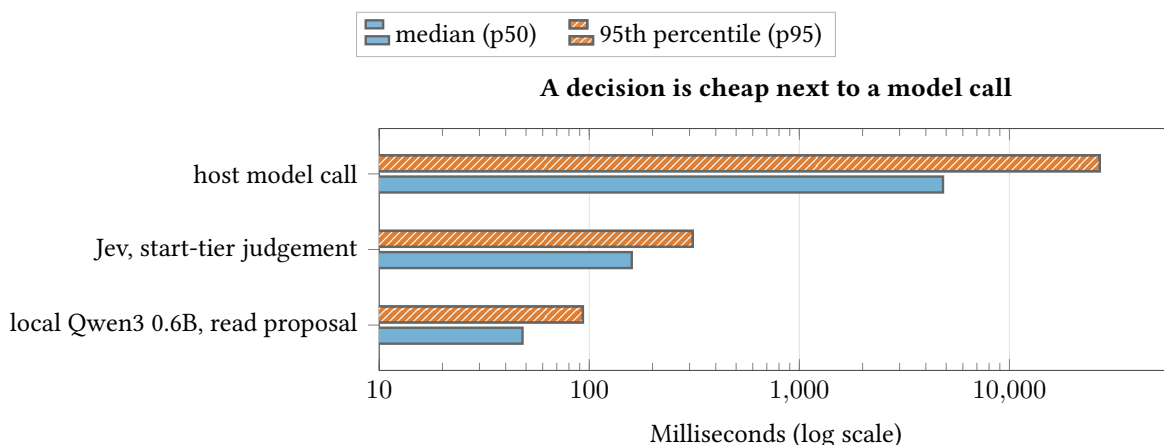


Figure 1: Latency of decision calls and of host model calls, measured by the bundle’s recorder (log scale).

How to read this figure. Each row has a median bar and a 95th-percentile bar. The decision rows are two orders of magnitude shorter than a host model call; the p95 bars show the slow tail.

- Where a match was possible, the real local scorer agreed with the host 77.5 % of the time (117 of 151; 95 % CI 70.2–83.4 %).

Agreement is still not correctness: the store keeps no state text, so whether a disagreement was a mistake cannot be judged from it.

Claimed savings were estimates. The bundle’s own receipts claimed \$222.66 saved by keeping caches warm (194 production receipts) and showed per-step routing to Sonnet costing money (\$−1.80). The read shortcut “would have” avoided 117 host turns in 994 decisions. None of these was measured as a saving; later measurement turned the per-step routing off. In production, the first start-tier decision of a session was most often the scope gate keeping the host (105 of 214 sessions, 49 %); Jev judged 68 sessions strong and 35 cheap.

Key idea

Telemetry told us decisions are cheap and fast, and that the bundle’s own savings claims were estimates. It could not tell us whether decisions were right or whether they saved money. That needed labelled cases and paired sessions.

3 Grading the decisions

To know whether a decision is right we need labelled cases. We built two benchmarks and graded every decision model under the gate the bundle ships: act only when the top probability is at least 0.90 and leads the runner-up by at least 0.20, within 3 s; otherwise fall back to the host.

Constructed cases. 10 judges answered 90 development cases (a screen) and 63 fresh holdout cases whose analysis was preregistered, 3 times each: file reads, searches, selections and computer-use questions, including cases with side effects or prompt injections. Two blind reviewers checked the labels (Cohen’s $\kappa = 1.00$ with the final labels). On the holdout, Jev was right on 55 of 63 cases at \$18.41 per million decisions and a p95 of 222 ms. GPT-6 Luna (60) and GPT-6.1 Sol (62) were slightly more accurate, but neither difference survived the preregistered Holm correction (adjusted $p = 0.16$ and 0.09), and both were slower and dearer (p95 1,850 and 4,252 ms). Cloudflare’s Clef, tested later and labelled post hoc, matched Jev’s accuracy (57 against an in-run Jev 56) at $3.2\text{--}4.4\times$ the cost per decision and a p95 of 663–738 ms from our machine.

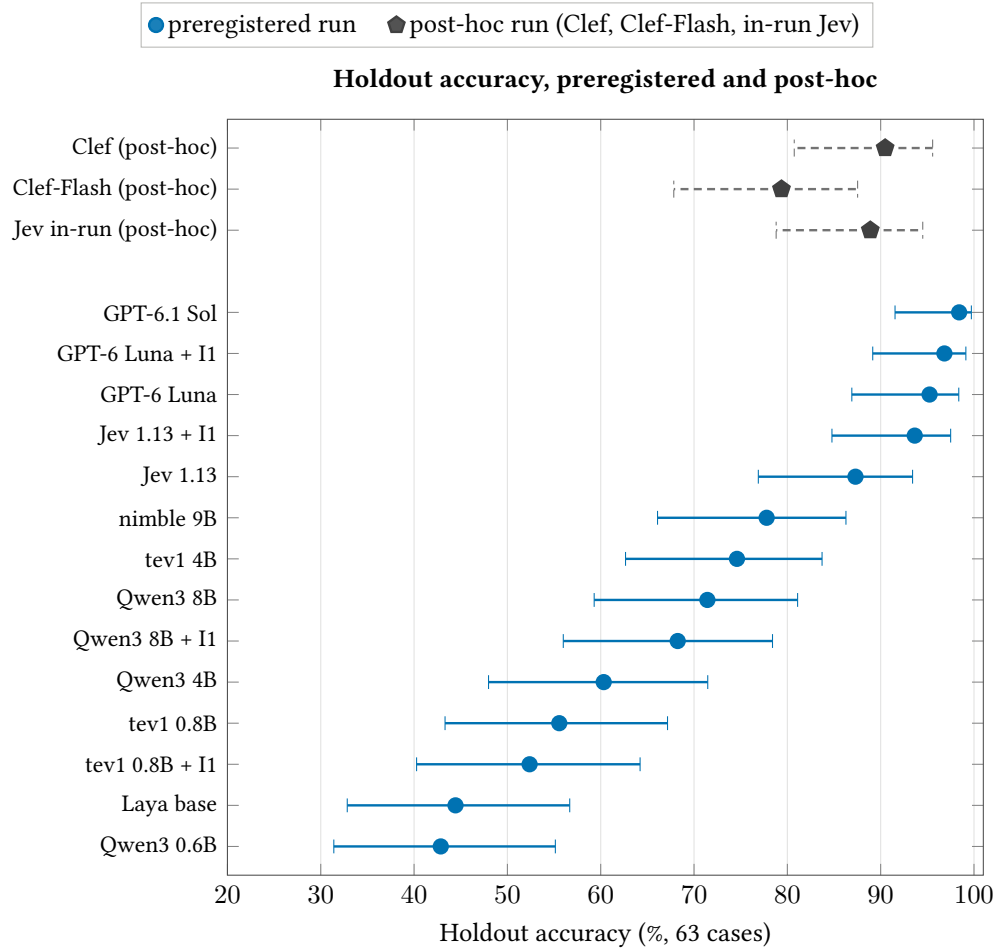


Figure 2: Holdout accuracy with 95 % Wilson intervals: the preregistered run, then (below the gap) the post-hoc run of Clef, Clef-Flash and the in-run Jev reference.

How to read this figure. Each row is a judge; the dot is its accuracy and the whisker its 95 % interval. The top judges' whiskers overlap, which is why their differences are called "within noise". Jev sits among them at a fraction of the cost and latency of the cloud models (companion report, Part I).

Real decisions. Constructed cases can be easier or harder than real ones. A second preregistered study rebuilt 42 real read-shortcut decisions from earlier sessions, exactly as the in-session judge had seen them. Reviewers agreed with each other ($\kappa = 0.84$) much more than with what the host actually did next ($\kappa = 0.51 - -0.57$). The preregistered usefulness rule asked for at least 4 correct automatic reads with few misfires. 0 of 10 judges met it (Fig. 3). Jev took the most correct reads (10 of 13) but also made 5 wrong automatic reads.

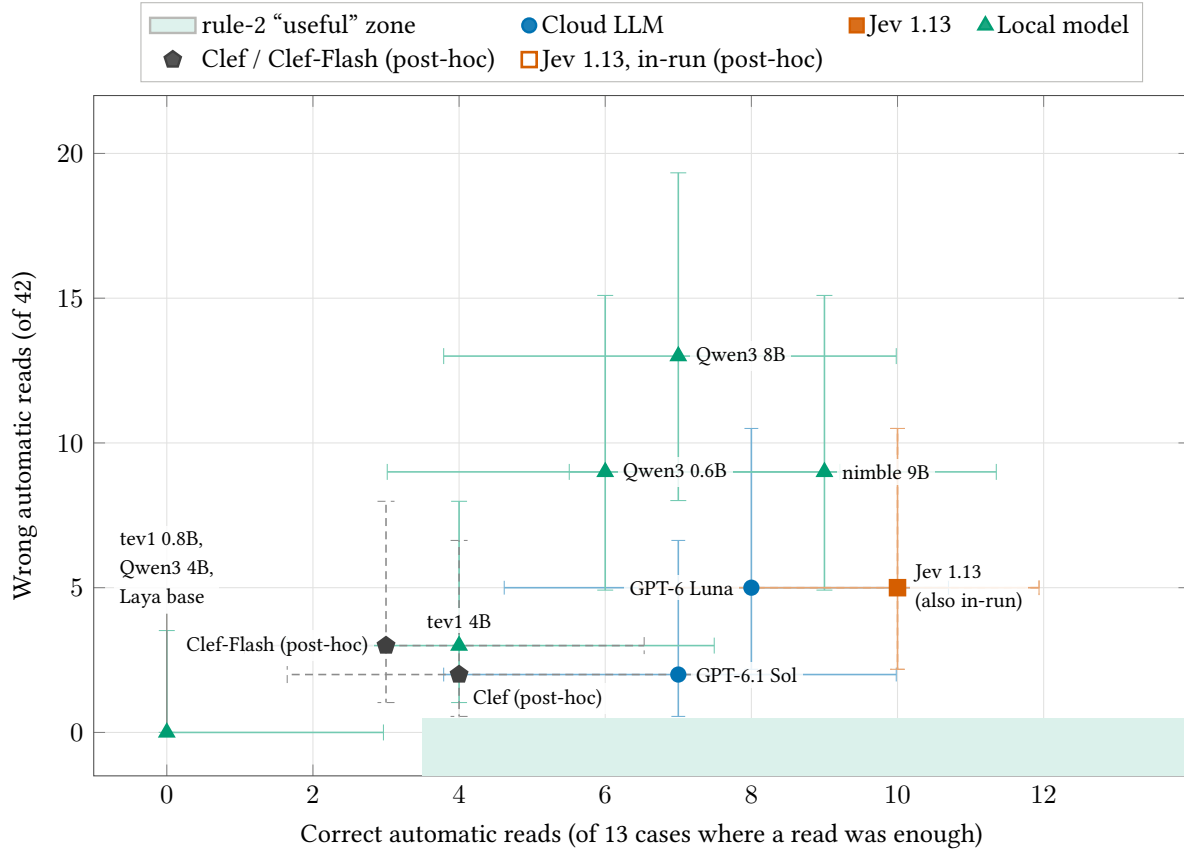


Figure 3: Real decisions: correct automatic reads against wrong automatic reads on trace-holdout, with 95 % Wilson intervals: the preregistered run, plus the post-hoc Clef arms (black pentagons). The shaded strip is where the preregistered rule would call a judge useful.

How to read this figure. Right is more saved host calls; up is more misfires. No judge reaches the shaded strip: every judge with enough correct reads also misfired.

Key idea

Jev is the best value among the decision models we tested: as accurate as the expensive ones within noise, and much cheaper and faster. But on real read decisions no model was safe enough to act on its own, so the read shortcut stays off. The money, if any, had to come from a bigger decision: which model runs the session.

4 Raising the difficulty: whole sessions

The decision that moves money is the session’s starting model. To measure it we ran whole multi-turn sessions, in pairs: the same scripted task, started together from the same frozen workspace, once on the plain host and once under a routing policy, each with its own cache key. The main campaign (main-v1) ran 1,036 sessions on 70 scenarios of 8 to 16 turns, with Fable 5.1 and Opus 5.5 as hosts. Hypotheses, thresholds and the split were

preregistered; the estimator was written after the data, which the companion report states plainly.

Why caching made this necessary. One of us (DK) asked whether routing still saves money in longer chats. Prompt caching makes re-sending a conversation cheap on the model that cached it, and switching models throws that cache away. An earlier survey had found the shipped router’s Opus-host ratio going from $0.95\times$ in single-turn sessions to $1.31\times$ in 4-turn ones. Estimates from receipts could not settle it.

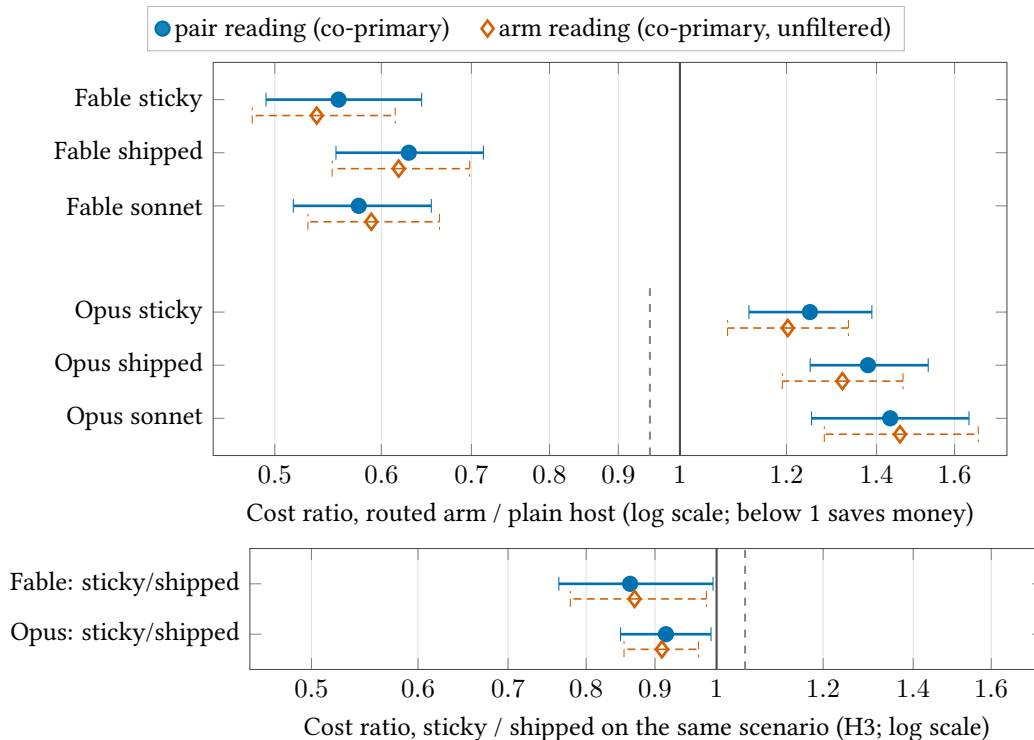


Figure 4: Confirmatory cost ratios on the main-v1 test split. Top: each routed arm against the plain host. Bottom: the sticky arm against the shipped arm on the same scenarios. Whiskers are 95 % intervals; the dashed lines mark preregistered thresholds.

How to read this figure. The dot is the geometric-mean cost ratio, the whisker its 95 % interval, on a log axis. Rows left of 1 saved money. All Fable rows are well left of 1; all Opus rows are right of it.

What it found. With Fable as host, choosing Sonnet once at the start of the session (*sticky*) cost $0.56\times$ the plain host (95 % CI 0.49–0.64; a typical per-session saving of 44 % as a geometric mean, and an actual spend cut of 37.3 % over all scenarios). Quality stayed inside the preregistered margin on the test split only; over all scenarios the margin was not shown (-0.021 , CI -0.058 to 0.012), and 3 task types lost more (review -0.155 , explain -0.143 , feature -0.071). With Opus as host, every routing arm cost more ($1.25\text{--}1.43\times$): Opus’s cache reads are cheap enough that leaving them costs more than Sonnet saves. That conclusion depends on the price table; Opus routing would break even if its cache reads cost about \$0.38 per million tokens instead of \$0.20.

The effort confound. An external peer reviewer noticed that routed Sonnet ran at medium reasoning effort while the plain-Sonnet control ran at the provider default. Two follow-ups measured effort directly. Plain Sonnet at medium cost $0.821\times$ its default ($0.781\text{--}0.861$; designed in advance, provenance-limited), and plain Fable at medium cost $0.860\times$ its default ($0.833\text{--}0.885$; preregistered), both with no detectable turn-pass loss on their 23 scenarios. So about 26 % of the Fable routing saving is the host’s own effort effect, and routing to Sonnet remains much the larger lever: sticky-on-Sonnet cost about $0.65\times$ what Fable at medium did (exploratory, not concurrent). The two review rounds and their responses are recorded with the companion report.

Key idea

Measured, not estimated: on Fable, deciding once to run the session on Sonnet saved about half the cost, with a quality caveat on some task types. On Opus it cost more. Lowering reasoning effort is a second, smaller lever that works on both Sonnet and Fable.

5 The plateau: what the decision model is actually worth

By now the open question was no longer which decision model is best, but whether the session-start decision needs a model at all, and how to configure it.

A counterfactual on the existing sessions. A decide-once policy makes one choice per session, so every policy’s cost can be read off the paired sessions already run: the plain host where it keeps the host, the routed session where it routes. [Figure 5](#) and [Table 1](#) price every such policy on all main-v1 scenarios. On Fable, Jev routed 89 % of sessions and cost $0.545\times$ the plain host. The simple rule R^* (route unless the workspace has more than 300 files) routed 99 % and cost $0.509\times$ at the same turn-pass difference (-0.019). Jev’s “keep on host” calls bought nothing: in every scenario-rep where Jev kept the host, the routed session scored the same. Jev and R^* disagreed on only 14 of 140 scenario-reps, which bounds how much they can differ. On Opus the price gate, which never routes there, sat at the oracle (regret \$1 per 1,000 sessions), while ungated Jev would have cost $1.159\times$.

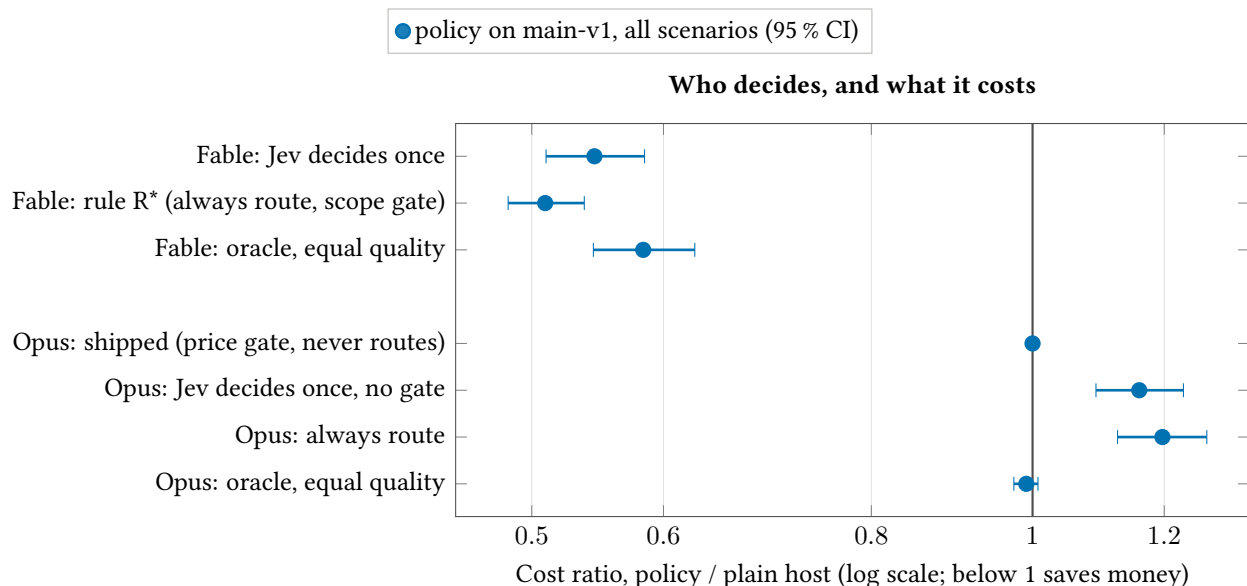


Figure 5: Decide-once policies priced on the existing main-v1 sessions (all scenarios, both reps), with 95 % scenario-cluster intervals. The oracle chooses with hindsight on the other rep, at equal quality. Exploratory.

How to read this figure. Each row is a policy. Left of 1 saves money. On Fable, the rule R^* is at least as cheap as Jev; on Opus, every policy that routes costs more than keeping the host.

An artefact to avoid. Sessions that changed reasoning effort request by request rewrote the prompt cache each time: within a turn, a request after an effort change wrote 11,650 tokens to the cache on average against 900 when effort stayed the same ($12.9\times$). Effort should be set once per session, like the model.

The core study (S1) [preregistered]. These findings were frozen into candidate configurations and a preregistered holdout study on 60 scenarios no agent had run: does the frozen configuration hold up, does the

Table 1: Decide-once policies on main-v1 (exploratory): share routed, geometric-mean cost ratio and turn-pass difference against the plain host, and dollars per 1,000 sessions.

Policy (decided once per session)	routed	cost ratio [95 % CI]	turn-pass Δ [95 % CI]	\$ / 1,000
Fable: always host (plain)	0 %	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	5,311
Fable: Jev decides once	89 %	0.545 [0.510, 0.585]	−0.019 [−0.052, 0.010]	3,108
Fable: rule R* (always route, scope gate)	99 %	0.509 [0.484, 0.538]	−0.019 [−0.052, 0.010]	2,739
Fable: oracle, equal quality	81 %	0.583 [0.544, 0.627]	0.012 [−0.004, 0.030]	3,268
Opus: shipped (price gate, never routes)	0 %	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	2,125
Opus: Jev decides once, no gate	89 %	1.159 [1.092, 1.232]	−0.009 [−0.033, 0.013]	2,498
Opus: always route	100 %	1.197 [1.125, 1.273]	−0.005 [−0.030, 0.019]	2,620
Opus: oracle, equal quality	22 %	0.991 [0.975, 1.008]	0.000 [−0.011, 0.009]	2,124

decision model beat the rule, and does medium effort help Opus? A deterministic freeze rule then picks each host’s configuration. S1 ran 1,224 sessions on 60 scenarios. Table 2 lists the verdicts; in plain words:

Table 2: S1 (holdout-v3) preregistered hypotheses. Cost ratios are geometric means over scenarios; Holm within the two preregistered families. H3b is descriptive.

Hypothesis	cost ratio (95 % CI)	turn-pass Δ (95 % CI)	Holm p	verdict
H1	0.581 (0.542–0.625)	−0.013 (−0.028–0.002)	0.000	✓ supported
H2	1.008 (1.000–1.022)	−0.003 (−0.008–0.000)	0.000	✓ supported
H3	0.999 (0.982–1.021)	0.007 (−0.004–0.022)	0.000	✓ supported
H3b	1.004 (0.987–1.022)	0.004 (−0.002–0.010)	–	descriptive
H4	0.992 (0.974–1.009)	−0.002 (−0.013–0.009)	0.187	✗ not supported
H5	1.255 (1.197–1.321)	−0.004 (−0.024–0.017)	0.000	✓ supported
H6	1.014 (0.993–1.035)	0.004 (−0.011–0.019)	0.001	✓ supported
H7	0.616 (0.565–0.672)	−0.028 (−0.063–0.013)	0.120	✗ not supported

- **H1 (Fable value), supported.** The frozen Fable configuration cost $0.581 \times$ plain Fable (95 % CI 0.542–0.625) with turn-pass -0.013 (-0.028 to 0.002). The saving is real, but smaller than predicted: the frozen prediction from main-v1 was 0.508 (0.483 – 0.536), and the observed ratio lies above that interval.
- **H2 (decision model against a rule), supported.** Letting Jev choose the session’s model and letting the one-line rule R* choose it were equivalent: $1.008 \times$ (90 % CI 1.000 – 1.019), turn-pass -0.003 . They disagreed on only 4 of 120 scenario-reps. The preregistered decision follows: ship the simpler rule R*, with no external call and no consent needed; Jev becomes an opt-in decider.
- **H3 (bundle overhead), supported.** The shipped bundle on Opus cost $0.999 \times$ plain Opus (90 % CI 0.984 – 1.017): no overhead. On Fable, the bundle with the host pinned cost $1.004 \times$ (descriptive).
- **H4 (Opus at medium effort), not supported.** Opus at medium cost $0.992 \times$ its default (95 % CI 0.974 – 1.009): no detectable saving, unlike Sonnet ($0.821 \times$) and Fable ($0.860 \times$).
- **H5 (routing on Opus), supported.** Routing to Sonnet still lost on Opus: $1.255 \times$ (1.197 – 1.321), on fresh scenarios and without the effort-switch artefact.
- **H6 (live equals predicted), supported.** Live decide-once sessions cost what the decomposition predicted ($1.014 \times$, 90 % CI 0.996 – 1.032).
- **H7 (review and explain on Sonnet), not supported.** On the 19 review and explain scenarios, the turn-pass lower bound for Sonnet against the host was -0.063 , below the -0.05 margin. The preregistered rule therefore keeps these task types on the host (keep_on_host: [review, explain]).

What the freeze rule chose. On Fable the rule picked `always_route | strong default | scope off | keep none (0.528×)`, not the preregistered `rstar | strong medium | scope 300 | keep none`. That makes it *selected on holdout*: it must replicate in S2 before it ships. It also conflicts with H7, which keeps review and explain on the host. What ships for Fable until S2 replicates is therefore the preregistered configuration plus `keep_on_host: [review, explain]`, applied through a keyword proxy (Section 6). On Opus no candidate qualified, so Opus keeps the shipped defaults: no routing, default effort.

Key idea

The decision model’s start-of-session choice was matched by a one-line rule on these hosts and prices. Where the money is, routing on Fable and never on Opus, price arithmetic and a rule decide it. The value of a system-1 model is not in this decision.

How the run went. The study spent \$4,975.51. 8 Fable sessions of one scenario (bleach-sanitize-review) were served partly by Opus, most likely the provider’s fallback (an inference); the preregistered rule excluded them from cost, so Fable cost contrasts use 59 scenarios. The stop rule for gate failures was not followed to the letter, because the gate is evaluated when rows are extracted, not live; this is a disclosed deviation. A Forge outage made 135 of 376 wave attempts (36 %) fail on infrastructure; every failed wave was rerun whole. The budget cap was raised from \$5,000 to \$6,300 during the run, though the final spend stayed under the original cap, and concurrency changed several times. The A/A copies differed in turn-pass by -0.047 on Opus (11 pairs), so small quality differences cannot be interpreted. Rerunning the analysis without the flagged scenarios changed no verdict and no freeze choice.

6 What to configure

Key idea

What gets the most out of a decision model: decide once per session; let price arithmetic gate routing (Opus: never route; Fable: route to Sonnet at medium); use a deterministic rule where it matches the model, as it did for the session-start choice; set effort to medium where it helps (Sonnet, Fable, not Opus); keep review and explain work on the host. Where a decision model may still pay is in decisions whose right answer changes within a session; that is a hypothesis for S2–S4, not a finding.

Table 3 gives the settings we recommend, per host model, after S1.

Table 3: Recommended settings per host model at the companion report’s prices, after S1.

Host model	Session model	Reasoning effort	Evidence
Fable 5.1	Route to Sonnet, decided once by rule R^* (scope gate at 300 files); review- and explain-like first prompts kept on the host via a keyword proxy	Sonnet at medium; Fable at medium when kept ($0.860\times$)	expected about $0.75\times$, turn-pass about 0 (replay, exploratory); R^* alone $0.581\times$ (S1 H1)
Opus 5.5	Keep the host: the price gate never routes (routing $1.255\times$)	Default (medium $0.992\times$, no saving)	S1 H3, H4, H5
Sonnet 5	Already the cheap model: no routing	Medium ($0.821\times$ its default)	Sonnet follow-up

What a user should set.

- **Price gate on.** The gate compares the host's and Sonnet's prices with the request mix measured in main-v1 (request multipliers 1.11 for Fable and 1.38 for Opus, re-derived by a replay of the shipped code, which passed) and keeps the host when routing cannot pay.
- **Decide once.** One model and one effort per session; switching either rewrites the prompt cache.
- **Effort medium on Sonnet and Fable,** default on Opus.
- **Decider: the rule R*.** It matched Jev on the session-start choice (H2) and needs no external call. Jev stays available as an opt-in decider; it sends the session's starting state to an external service and should run only with the user's consent (`allow_external_state`).
- **keep_on_host: [review, explain]** on Fable, per H7: a deliberate quality-first choice. Without a judge there is no task-type label, so the shipped bundle uses a frozen keyword classifier (`intent-kw-v1`) as a proxy: it keeps sessions whose first prompt reads as a question on the host. On the 135 scenario prompts of all three panels it catches 26 of the 28 review and explain sessions, but also 43 other read-only questions (precision 38 %); it was not tuned on the holdout. Replayed on the S1 sessions (exploratory, not preregistered), against plain Fable:
 - R* alone: $0.581\times$, turn-pass -0.013 (the preregistered H1 result);
 - R* with review and explain kept on the host by their true labels: $0.650\times$ ($0.601-0.703$), turn-pass -0.006 ;
 - R* with the shipped keyword proxy: $0.749\times$ ($0.699-0.799$), turn-pass 0.002 .

The proxy gives up part of the saving (about $0.75\times$ instead of about $0.58\times$) to avoid the review and explain quality risk. Users who accept that risk can remove `keep_on_host` to get about $0.58\times$. A better classifier is S2 work.

Per harness. In Amplifier, the fast-decisions bundle makes the decision inside every session; install it with `amplifier bundle add` as in the repository README and keep it current with `amplifier bundle update`. In Claude Code, Codex or Copilot CLI, the smart tool makes the same decision before the harness starts. Its `decide` command returns the model and effort for a task; `launch` decides and then starts the harness with them:

```
amplifier-fast-decisions decide    amplifier-fast-decisions launch
```

Install and update the skill with:

```
npx skills add michaeljabbour/amplifier-bundle-fast-decisions
npx skills update
```

7 What we did not finish

The plan for this program (`docs/design/v3/PLAN.md`) has five sessions studies; only the first is in this report. Each of the others has a written design there:

- **S2, long sessions:** does decide-once still hold when a session runs far beyond 16 turns?
- **S3, delegation in situ:** routing the sub-tasks an agent delegates, inside real sessions.

- **S4, other harnesses:** the measured saving of decide and launch in Claude Code, Codex and Copilot CLI.
- **S5, SWE-bench at full scale:** two arms on the full benchmark.
- **Replicating the Fable freeze choice in S2** (it was selected on holdout), and anything on hosts other than Fable, Opus and Sonnet.

Honest limits. Everything here ran on one provider, at one price table: costs are list prices applied to measured tokens, not invoices, and the Opus conclusion depends on its cache-read price. The sessions are scripted, at most 16 turns, on public code; the counterfactual reuses main-v1's sessions and its cheap outcomes for a few scenario-reps are estimates. The telemetry is one owner's workload. Decision quality on real reads remains unidentified beyond the 42 labelled cases. The judge's own cost is small but was estimated, not billed. S1 had operational deviations ([Section 5](#)): a raised budget cap, a stop rule not followed to the letter, and many infrastructure reruns.

A Companion material

Companion technical report. The full studies behind [Sections 3](#) and [4](#) (design, power, running safely, every confirmatory and exploratory result, the effort follow-ups, three rounds of review, data dictionary and reproduction) are in docs/papers/2026-10-02-paired-measurement/. The external reviews and their verification are in docs/reviews/2026-10-paired-measurement/.

Evidence used here.

- Observatory and counterfactual: docs/evidence/2026-10-v3-offline/ (A1, A0, A3 inputs).
- Judge benchmark, real traces, Clef:
docs/evidence/2026-09-30-judge-benchmark/
docs/evidence/2026-10-01-trace-judge-benchmark/
docs/evidence/2026-10-04-clef-judges/
- Paired campaign and follow-ups:
docs/evidence/2026-10-02-paired-campaign/
docs/evidence/2026-10-05-effort-control/
docs/evidence/2026-10-06-effort-control-fable/
docs/evidence/2026-10-05-defaults-replay/
- S1:
docs/evidence/2026-10-06-holdout-v3/
preregistration: evals/paired/PREREGISTRATION-holdout-v3.md

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